Analyse methods, conclusions and claims for assumptions, possible sources of error, conflicting evidence and unanswered questions

Learning intention

- analyse methods, conclusions and claims for assumptions, possible sources of error, conflicting evidence and unanswered questions.

Teach and model

- identifying sources of error in methods such as inconsistent variable control and inaccuracies in procedures or measurements.
- identifying assumptions then examining if extra variable controls are required and how these might affect the data and conclusion.
- analysing conclusions or claims to determine if there are further questions which should be explored to verify the conclusion or claim.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.